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Course: CSA - 1304 Theory of Computation

CO1 - Assessment Tool-1

Given:  $Q = \{q_0, q_1, q_2\}$   
 $\Sigma = \{a, b\}$

Start state:  $q_0$

Final state:  $q_2$  (F)

NFA Transition table

| State             | a              | b              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_2\}$      |
| $q_1$             | $\{q_0\}$      | $\{q_1\}$      |
| $\star q_2$       | $\emptyset$    | $\{q_0, q_1\}$ |

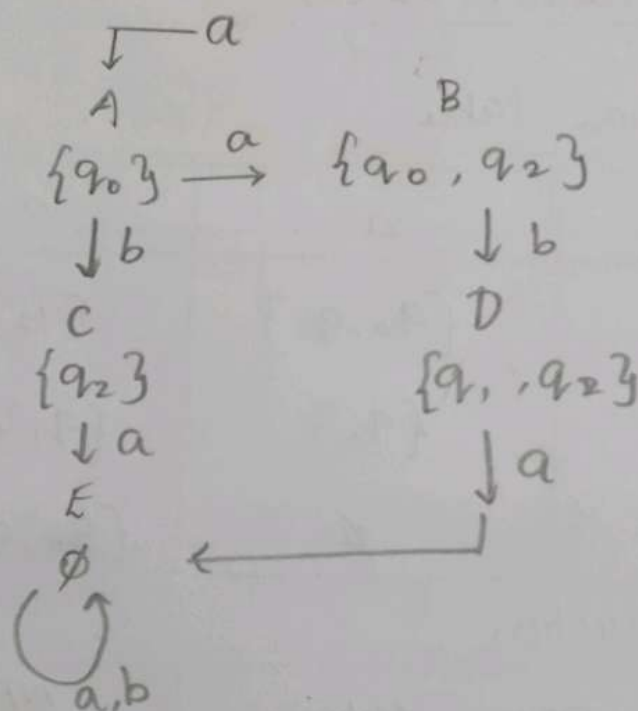
Subset construction

| DFA States | NFA States     |
|------------|----------------|
| A          | $\{q_0\}$      |
| B          | $\{q_0, q_1\}$ |
| C          | $\{q_2\}$      |
| D          | $\{q_1, q_2\}$ |
| E          | $\emptyset$    |

Equivalent DFA transition table:

| DFA | State | a | b |
|-----|-------|---|---|
| →   | A     | B | c |
|     | B     | B | D |
| *   | C     | E | B |
| *   | D     | A | B |
|     | E     | E | E |

Equivalent DFA diagram:



Equivalent DFA

| State | a | b |
|-------|---|---|
| → A   | B | c |
| B     | B | D |
| * C   | E | B |
| * D   | A | B |
| E     | E | E |

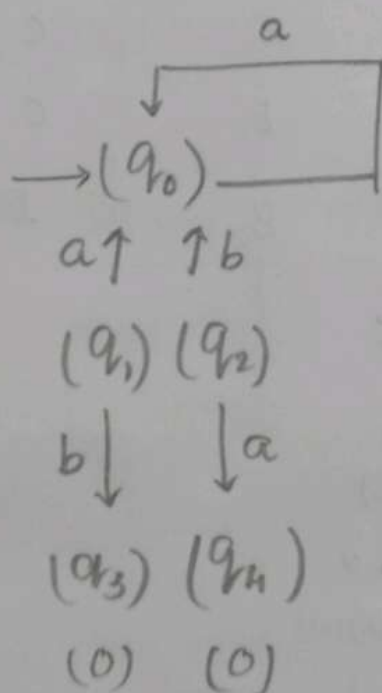
2). Given :  $\Sigma = \{a, b\}$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

NFA Transition Table

| State             | a              | b              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_2\}$ |
| $q_1$             | -              | $\{q_3\}$      |
| $q_2$             | $\{q_4\}$      | -              |
| $\star q_3$       | -              | -              |
| $\star q_4$       | -              | -              |

NFA diagram :



## Equivalent DFA

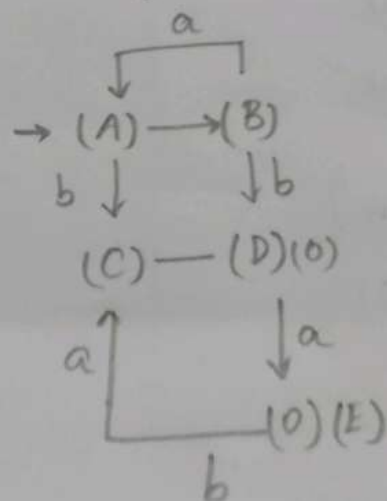
Subset Construction:

| DFA States | NFA States          |
|------------|---------------------|
| A          | $\{q_0\}$           |
| B          | $\{q_0, q_1\}$      |
| C          | $\{q_0, q_2\}$      |
| D          | $\{q_0, q_1, q_2\}$ |
| E          | $\{q_0, q_1, q_4\}$ |

DFA transition table

| State           | a | b |
|-----------------|---|---|
| $\rightarrow$ A | B | C |
| B               | B | D |
| C               | E | C |
| * D             | E | C |
| * E             | B | D |

Equivalent DFA diagram:



### Equivalent DFA

| State           | a | b |
|-----------------|---|---|
| $\rightarrow A$ | B | C |
| B               | B | D |
| C               | E | C |
| $\star D$       | E | C |
| $\star E$       | B | D |

3. Given:

$$\Sigma = \{0, 1\}$$

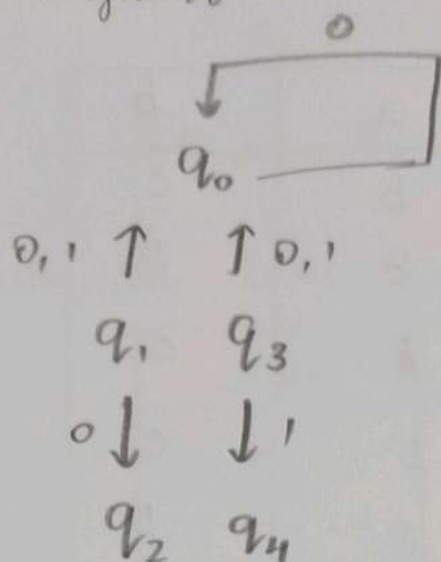
$$M = (Q, \Sigma, \delta, q_0, F)$$

$$F = \{q_2, q_4\}$$

NFA transition table

| State             | 0              | 1              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_3\}$ |
| $q_1$             | $\{q_2\}$      | -              |
| $q_2$             | -              | -              |
| $q_3$             | -              | $\{q_4\}$      |
| $q_4$             | -              | -              |

NFA diagram:



String verification : input = 01011

| input Symbol | Current State |
|--------------|---------------|
| Start        | $q_0$         |
| 0            | $q_0$         |
| 1            | $q_0$         |
| 0            | $q_0$         |
| 1            | $q_0$         |
| 1            | $q_0$         |

$$M = (Q, \Sigma, \delta, q_0, F)$$

Transition table:

| State             | 0              | 1              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_3\}$ |
| $q_1$             | $\{q_2\}$      | -              |
| $\star q_2$       | -              | -              |
| $q_3$             | -              | $\{q_4\}$      |
| $\star q_4$       | -              | -              |

4) Given:

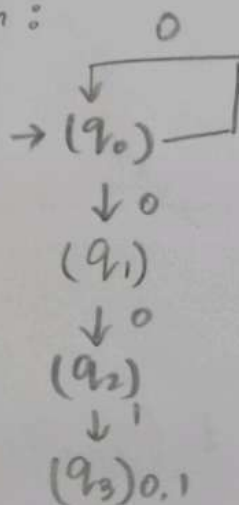
$$\Sigma = \{0, 1\}$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

NFA transition table

| State             | 0              | 1         |
|-------------------|----------------|-----------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0\}$ |
| $q_1$             | $\{q_2\}$      | -         |
| $q_2$             | -              | $\{q_3\}$ |
| $\star q_3$       | $\{q_3\}$      | $\{q_3\}$ |

NFA diagram:



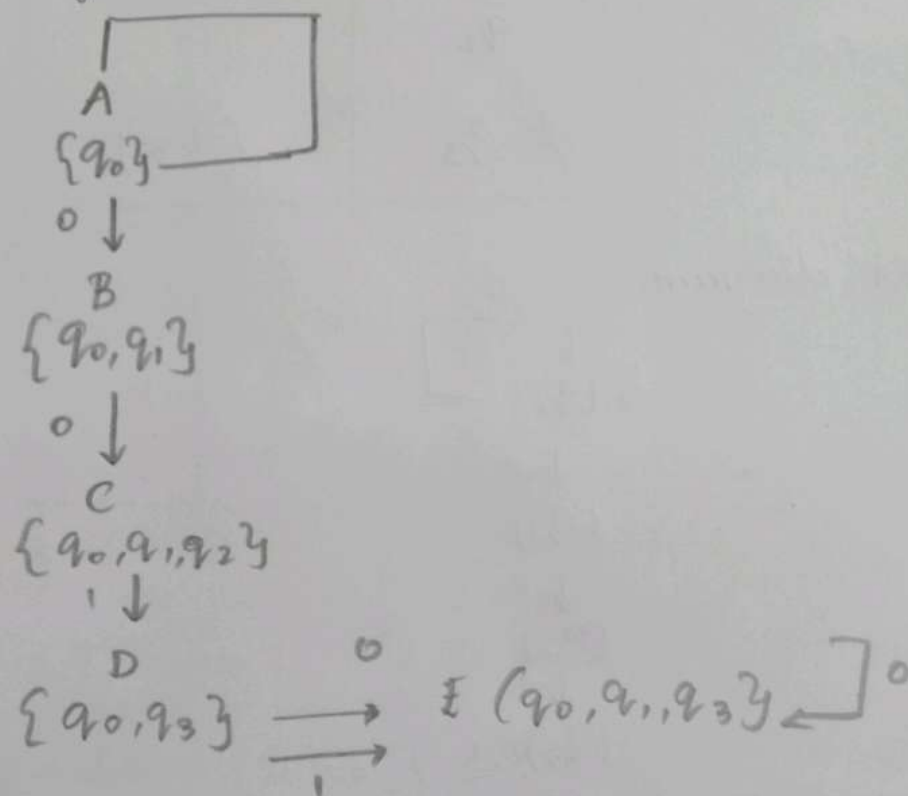
Subset construction

| DFA State | NFA State           |
|-----------|---------------------|
| A         | $\{q_0\}$           |
| B         | $\{q_0, q_1\}$      |
| C         | $\{q_0, q_1, q_2\}$ |
| D         | $\{q_0, q_3\}$      |
| E         | $\{q_0, q_1, q_3\}$ |

Equivalent DFA transition table

| State | 0 | 1 |
|-------|---|---|
| → A   | B | A |
| B     | C | A |
| C     | C | D |
| * D   | E | D |
| * E   | E | D |

Equivalent DFA diagram:



Equivalent DFA

| State | 0 | 1 |
|-------|---|---|
| → A   | B | A |
| B     | C | A |
| C     | C | D |
| * D   | E | D |
| * E   | E | D |

5) Given:  $Q = \{q_0, q_1, q_2\}$   
 $\Sigma = \{0, 1\}$

Start State :  $q_0$

Final State :  $q_2 \{F\}$

$\epsilon$  transitions

$$q_0 \xrightarrow{\epsilon} q_1$$

$$q_1 \xrightarrow{\epsilon} q_2$$

Transitions

| State             | 0     | 1     |
|-------------------|-------|-------|
| $\rightarrow q_0$ | -     | $q_0$ |
| $q_1$             | $q_1$ | -     |
| $* q_2$           | $q_2$ | $q_2$ |

$\epsilon$ -closure ( $q_0$ )

From  $q_0$  :  $q_0 \xrightarrow{\epsilon} q_1 \xrightarrow{\epsilon} q_2$

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$\epsilon$ -closure ( $q_1$ )

From  $q_1$  :  $q_1 \xrightarrow{\epsilon} q_2$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$\epsilon$ -closure ( $q_2$ )

No  $\epsilon$ -transition from  $q_2$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

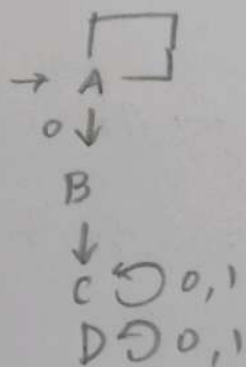
DFA States

| DFA State | NFA State           |
|-----------|---------------------|
| A         | $\{q_0, q_1, q_2\}$ |
| B         | $\{q_1, q_2\}$      |
| C         | $\{q_2\}$           |
| D         | $\emptyset$         |

DFA Transition table

| State             | 0 | 1 |
|-------------------|---|---|
| $\rightarrow$ * A | B | A |
| * B               | B | C |
| * C               | C | C |
| D                 | D | D |

Equivalent DFA diagram:



$\epsilon$ -closure Table

| State | $\epsilon$ -closure |
|-------|---------------------|
| $q_0$ | $\{q_0, q_1, q_2\}$ |
| $q_1$ | $\{q_1, q_2\}$      |
| $q_2$ | $\{q_2\}$           |